

Connected minimal dominating sets of trapezohedral graphs

A proof of the generating-function conjecture for OEIS A381190

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Abstract

Let a_n count the vertex subsets of the n -trapezohedral graph that are inclusion-minimal dominating sets and induce connected subgraphs. We prove the rational generating function conjectured by Joerg Arndt in OEIS A381190. The proof classifies every such set: it contains exactly one of the two poles and exactly one vertex of the opposite rim type. The remaining choices form a cyclic binary word built from the blocks 01 and 011, together with a restricted marked edge. A marked-block bijection gives

$$a_n = 2n[x^n] \frac{2x^2 + x^3}{1 - x^2 - x^3}, \quad n \geq 3.$$

We derive a finite binomial-sum formula, a bivariate generating function recording set size, two recurrences, a three-root formula, and a sharp exponential asymptotic. A nearest-integer formula is proved for every $n \geq 45$. Independent exhaustive enumeration verifies the classification as an equality of families of subsets through $n = 10$; exact arithmetic checks all published terms and the derived identities through $n = 1000$. The archive records the four-candidate shortlist and the single random draw used to select this problem.

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1 The conjecture and the result

OEIS A381190 counts connected minimal dominating sets in the n -trapezohedral graph, beginning with $n = 3$. Its initial values are

$$6, 16, 30, 36, 70, 96, 144, 220, 308, 456, 650, 924, \dots \quad (1.1)$$

The entry attributes the sequence to Eric W. Weisstein, its extension from $n = 14$ onward to Christian Sievers, and the following generating function conjecture to Joerg Arndt, January 7, 2026 [1]:

$$A(x) = \sum_{n \geq 3} a_n x^n = \frac{-2x^3(4x^5 + 8x^4 + 4x^3 - 9x^2 - 8x - 3)}{(x^3 + x^2 - 1)^2}. \quad (1.2)$$

The retrieved internal entry, revision 16 dated January 7, 2026, still labels this formula conjectural. The retrieval date for the present article is September 20, 2026.

Theorem 1.1 (Main result). *The generating function in (1.2) is correct. More precisely, for every $n \geq 3$,*

$$a_n = 2n[x^n] \frac{2x^2 + x^3}{1 - x^2 - x^3}. \quad (1.3)$$

If $p_m = 0$ for $m < 0$ and

$$\sum_{m \geq 0} p_m x^m = \frac{1}{1 - x^2 - x^3}, \quad (1.4)$$

then the following general-term formulas also hold:

$$a_n = 2n(2p_{n-2} + p_{n-3}), \quad (1.5)$$

$$a_n = 2n \sum_{\substack{r, s \geq 0 \\ 2r + 3s = n}} \left(2 \binom{r + s - 1}{s} + \binom{r + s - 1}{s - 1} \right). \quad (1.6)$$

Here a binomial coefficient is zero when its lower index lies outside its usual range.

This is not an identity obtained by interpolating the displayed terms. The essential step is a classification of all the counted vertex sets, including a converse that supplies private neighbors witnessing their minimality. The generating function follows from a bijection valid for all n .

The problem concerns a natural polyhedral graph family, with the cube as its first member [2]. Its three simultaneous conditions—domination, inclusion-minimality, and induced connectedness—are sufficiently restrictive to admit a short encoding, but none can simply be omitted. In particular, the terminology needs care: a *connected minimal dominating set* is not the same object as a *minimal connected dominating set*.

1.1 Scope and provenance

The target was chosen by one random draw from four explicitly conjectural formulas. The ordered shortlist was fixed before that draw; candidate 4, A381190, was selected, and there were no rerolls. [Section A](#) gives the audit details. The other three conjectures are not asserted to have been proved here.

The result established in this note is a complete proof of the displayed identity under the stated graph and minimality conventions. The OEIS status was checked, but an exhaustive priority search was not performed; no claim of being the first proof anywhere in the literature is made. The computational material is independently reproducible, but is not a formal proof-assistant verification or a substitute for peer review.

2 Graph model and minimality

2.1 A precise model of the graph

For $n \geq 3$, let T_n have vertices

$$\{u, v\} \cup \{a_1, \dots, a_n\} \cup \{b_1, \dots, b_n\}$$

and edges

$$ua_i, \quad vb_i, \quad a_ib_i, \quad b_ia_{i+1} \quad (1 \leq i \leq n), \quad (2.1)$$

where $a_{n+1} = a_1$. The vertices u, v are called *poles*; the remaining vertices form the *rim*

$$a_1, b_1, a_2, b_2, \dots, a_n, b_n, a_1.$$

This is the standard trapezohedral graph model; the vertex and edge counts are $2n + 2$ and $4n$, respectively, and T_3 is the cubical graph [2]. Equations (2.1), rather than a drawing, are the complete specification used in both the proof and the verifier.

A useful automorphism is rotation by one rim vertex:

$$u \longleftrightarrow v, \quad a_i \mapsto b_i, \quad b_i \mapsto a_{i+1}. \quad (2.2)$$

Each kind of edge in (2.1) maps to another edge, so the map is indeed an automorphism. It will account for a factor of two.

Throughout, we count actual subsets of the vertex set of this fixed graph. Rotated or reflected subsets are *not* identified. There is no division by the automorphism group of T_n .

Definition 2.1. Write $N[w]$ for the closed neighborhood of a vertex w . A subset D of the vertices is *dominating* if $N[w] \cap D \neq \emptyset$ for every vertex w . It is *minimal dominating* if no proper subset of D is dominating. It is counted by a_n if, in addition, the induced graph $T_n[D]$ is connected.

Minimal means inclusion-minimal, not minimum cardinality. Moreover, minimality is tested among *all* dominating subsets, not just connected ones. Deleting a vertex and retaining domination disproves minimality even when the remaining induced subgraph is disconnected. This distinction is explicitly made in the source definition linked from the OEIS entry [3].

2.2 Private neighbors

Lemma 2.2 (Private-neighbor criterion). *A dominating set D is minimal dominating if and only if, for each $d \in D$, there exists a vertex w such that*

$$N[w] \cap D = \{d\}. \quad (2.3)$$

Such a vertex is called a private neighbor of d with respect to D . If $T_n[D]$ is connected and $|D| > 1$, every private neighbor lies outside D .

Proof. If no vertex satisfies (2.3) for d , every vertex remains dominated after deleting d , so D is not minimal. Conversely, if such a vertex exists for every d , deleting any one vertex destroys domination. If a proper subset D' were dominating, choose $d \in D \setminus D'$. Its private neighbor would be undominated by D' , a contradiction. This also explains why testing all single vertex deletions is sufficient.

Finally, if $w \in D$ and (2.3) holds, then $w = d$ because $w \in N[w]$. In a connected induced graph with more than one vertex, d has a neighbor in D , so $N[d] \cap D$ cannot be the singleton $\{d\}$. $\square$

No singleton dominates T_n : a pole misses the other pole and all rim vertices of the opposite type, while a rim vertex has degree three and T_n has at least eight vertices. Thus every set considered below has more than one vertex, and the last assertion of Lemma 2.2 applies.

3 A structural classification

3.1 Exactly one pole is selected

Lemma 3.1. *Every connected minimal dominating set of T_n contains exactly one of u and v .*

Proof. Suppose first that both poles belong to D . If $a_i \in D$, its possible private neighbors are a_i, u, b_{i-1}, b_i . The first two have u in their closed neighborhoods, and the last two have v . Consequently none can be private to a_i . The same argument, with the types exchanged, excludes every selected b_i . Hence the only remaining possibility is $D = \{u, v\}$, which is disconnected because uv is not an edge.

Suppose instead that neither pole is selected. Then D is a connected subset of the rim cycle, so it is either the entire rim or a consecutive path on that cycle. If it is the entire rim, each selected vertex has both rim neighbors selected, and its adjacent pole has at least two selected neighbors. No selected vertex has a private neighbor.

If the path contains at least four vertices, write its first four vertices in order as w_1, w_2, w_3, w_4 . The vertex w_2 has both rim neighbors in D . Its remaining neighbor is a pole also adjacent to w_4 , since w_2 and w_4 have the same rim type. Its closed neighborhood possibilities therefore provide no private neighbor, contradicting Lemma 2.2.

A path of at most three selected rim vertices dominates at most five rim vertices: the selected path and at most one additional vertex at each end. The rim has $2n \geq 6$ vertices, so such a set is not dominating. Both alternatives have been excluded. $\square$

By (2.2), the two pole choices contribute equally. We may therefore classify the sets satisfying $u \in D, v \notin D$, and multiply their number by two at the end.

3.2 The opposite rim contains a single selected vertex

Lemma 3.2. *If D is counted and $u \in D, v \notin D$, then exactly one of $b_1, \dots, b_n$ belongs to D .*

Proof. At least one b_i is selected in order to dominate v . For any selected b_i , its two neighboring a -vertices cannot be private neighbors, because each is also adjacent to the selected pole u . The vertex b_i cannot be its own private neighbor by Lemma 2.2. Thus its only possible private neighbor

is v . But v is private to b_i precisely when there is no other selected b -vertex. This proves the assertion. $\square$

Write this unique selected vertex as b_j . Encode the selected a -vertices by a cyclic word

$$x_1 x_2 \cdots x_n, \quad x_i = \mathbf{1}_{\{a_i \in D\}} \in \{0, 1\}. \quad (3.1)$$

Associate the selected b_j with a mark on the edge between positions j and $j + 1$ of the cyclic word. This is an edge of the compressed cycle of a -positions, not an additional edge of T_n .

Lemma 3.3. *The cyclic word in (3.1) contains neither 00 nor 111. Consequently it has a unique decomposition into cyclic blocks of the forms 01 and 011, each block starting at a zero.*

Proof. If $x_i = x_{i+1} = 0$ and $i \neq j$, then the unselected vertex b_i is undominated: its only neighbors are the absent pole v and the two absent a -vertices. If $i = j$, the selected vertex b_j has no neighbor in D , contradicting connectedness. Thus 00 is forbidden, including across the cyclic boundary.

A selected a_i can have a private neighbor only among its two adjacent b -vertices. Neither itself nor u can be private to it, because u is selected. An adjacent b -vertex can be private to a_i only if that b -vertex is not the selected b_j and its other a -neighbor is absent. In particular, a_i must have a zero next to its position in the compressed cyclic word. A run of three ones has a middle one without such a zero, so 111 is forbidden.

There is at least one zero, since an all-one word of length $n \geq 3$ contains 111. The zeros are separated by runs of one or two ones. Starting a block at each zero gives the asserted unique decomposition. $\square$

3.3 The legal positions of the mark

A singleton run of ones will mean the 1 in a block 01; a double run will mean the two 1's in a block 011. The distinction controls which adjacent b -vertex may be selected.

Lemma 3.4. *A mark is compatible with minimality exactly in the following cases:*

- (i) *it lies between the two ones of a double run;*
- (ii) *it lies on either edge incident to the one of a singleton run.*

Thus a block 011 supplies one possible mark, and a block 01 supplies two possible marks.

Proof. There are no 00 edges. If the marked edge is 11, those two ones form a double run. Each of them still has an unmarked edge toward a zero on its other side; the corresponding unselected b -vertex is a private neighbor. All other selected a -vertices are unaffected.

Suppose the marked edge has endpoints 0, 1. If that one belongs to a double run, its only adjacent zero is across the marked edge. The corresponding b_j cannot be its private neighbor because b_j is selected, and its other neighboring b -vertex has another selected a -neighbor. Minimality fails. If instead the one is a singleton, it has zeros on both sides. The unmarked edge on its other side supplies a private neighbor. The same reasoning applies to an edge with endpoints 1, 0. $\square$

The mark on the right side of a singleton may lie at the boundary between blocks. It is assigned to the unique block containing that singleton one. This convention will be important for the counting bijection.

Theorem 3.5 (Exact classification). *Fix $u \in D$, $v \notin D$. Connected minimal dominating sets of T_n are in bijection with cyclic binary words on the labeled positions $1, \dots, n$, decomposed into blocks 01 and 011, together with one of the legal marks in Lemma 3.4. The set consists of u , the a -vertices at positions occupied by ones, and the unique b -vertex at the marked edge.*

If the word has r blocks 01 and s blocks 011, then

$$n = 2r + 3s, \quad |D| = 2 + r + 2s, \quad \text{number of legal marks} = 2r + s. \quad (3.2)$$

Proof. The preceding lemmas prove necessity and uniqueness of the encoding. We verify sufficiency, including the private neighbor of the pole, which is not covered by the local restrictions alone.

Given a word and a legal mark, every a -vertex is dominated by u . Every b -vertex has at least one selected adjacent a -vertex because there is no 00. The pole v is dominated by the selected b_j . The induced subgraph is connected: every selected a -vertex is joined to u , and b_j is joined to at least one selected a -vertex.

The vertex v is private to b_j . Each selected a_i has an adjacent zero across an unmarked edge, by the mark rule. The unselected b -vertex on that edge has a_i as its only selected neighbor, so it is private to a_i .

It remains to find a private neighbor for u . An absent a -vertex at a zero position is private to u unless it is incident to the marked edge. If the mark joins two ones, every zero works. If the mark is next to a singleton one, only its zero endpoint is excluded. There must then be another zero: a cyclic word with only one zero and a singleton run would have length two, whereas $n \geq 3$. A different zero supplies the required private neighbor of u . Thus every selected vertex has a private neighbor and the set is minimal dominating.

The length, cardinality, and number of marks in (3.2) follow by summing the contributions of the two block types. The extra two selected vertices are u and b_j . $\square$

Example 3.6. For $n = 5$, take the cyclic word 01101 and mark the edge between its first two ones, corresponding to b_2 . Then

$$D = \{u, a_2, a_3, a_5, b_2\}.$$

A complete private-neighbor certificate is

Selected vertex	u	a_2	a_3	a_5	b_2
Private neighbor	a_1	b_1	b_3	b_4	v

The word has one block of each type. There are five distinct labeled cyclic words of this form, each with three legal marks. Including the two pole choices gives $a_5 = 2 \cdot 5 \cdot 3 = 30$.

4 The marked-block bijection and the generating function

4.1 A linear encoding of a cyclic object

Set

$$W(x) = \frac{2x^2 + x^3}{1 - x^2 - x^3}. \quad (4.1)$$

The numerator describes the distinguished block: a block 01 with one of its two marks, or a block 011 with its unique mark. The factor

$$\frac{1}{1 - x^2 - x^3} = \sum_{k \geq 0} (x^2 + x^3)^k$$

describes an arbitrary ordered list of further, unmarked blocks. All these are formal power series; analytic convergence is not needed for the counting argument.

A decorated cyclic word from [theorem 3.5](#) determines one distinguished block, namely the block carrying its mark. Start at that block's initial zero and read clockwise. This gives a marked first block followed by an ordinary linear list of blocks, exactly as counted by $W(x)$. Conversely, such a list of total length n , together with a choice of the labeled position of its initial zero, reconstructs the decorated cyclic word uniquely.

There are n choices of initial position. There is no ambiguity from periodic words: the mark already determines a specific block, and hence a specific zero at which to start. Rotating a list without also moving this distinguished block does not produce a second encoding of the same marked object. This avoids a potentially incorrect division by the number of blocks or by the rotation group.

For a fixed selected pole, the count is therefore $n[x^n]W(x)$. The pole-exchanging automorphism doubles it. This proves [\(1.3\)](#).

4.2 The small-degree correction

Multiplication of the coefficient of x^n by n is effected by the formal operator $x \frac{d}{dx}$. However, $W(x)$ also has the term $2x^2$, corresponding to the single block 01 of length two. The trapezohedral sequence starts at $n = 3$, and that word is precisely the exceptional case in which the pole need not have a private neighbor. We must remove its formal contribution:

$$A(x) = 2xW'(x) - 8x^2. \quad (4.2)$$

This is an index correction, not an assertion about a value a_2 of the stated graph sequence.

Put $Q(x) = 1 - x^2 - x^3$. Direct differentiation gives

$$\begin{aligned} 2xW'(x) &= \frac{2x((4x + 3x^2)Q(x) + (2x^2 + x^3)(2x + 3x^2))}{Q(x)^2} \\ &= \frac{8x^2 + 6x^3 + 2x^5}{Q(x)^2}. \end{aligned}$$

Since

$$Q(x)^2 = 1 - 2x^2 - 2x^3 + x^4 + 2x^5 + x^6,$$

subtracting $8x^2$ yields

$$A(x) = \frac{6x^3 + 16x^4 + 18x^5 - 8x^6 - 16x^7 - 8x^8}{1 - 2x^2 - 2x^3 + x^4 + 2x^5 + x^6}. \quad (4.3)$$

Factoring the numerator and using $Q(x)^2 = (x^3 + x^2 - 1)^2$ gives exactly [\(1.2\)](#). The conjectured generating function is proved.

The squared denominator has a transparent origin. The block list has denominator Q , and pointing to a labeled location introduces the factor n , equivalently a derivative. It is not evidence for an unexplained six-state recurrence guessed from data.

5 General-term formulas and recurrences

5.1 An elementary recurrence and a finite sum

The numbers p_m in (1.4) count compositions of m into parts two and three. They satisfy

$$p_0 = 1, \quad p_1 = 0, \quad p_2 = 1, \quad p_m = p_{m-2} + p_{m-3} \quad (m \geq 3), \quad (5.1)$$

with the negative-index convention already stated. This is a Padovan-type recurrence, with its indexing fixed explicitly here. The first values are

$$1, 0, 1, 1, 1, 2, 2, 3, 4, 5, 7, 9, 12, \dots$$

Multiplying its generating function by $2x^2 + x^3$ proves (1.5).

For a recurrence-free expression, fix r, s with $2r + 3s = n$ and write $k = r + s$. If the marked first block is 01, there are two mark choices and $\binom{k-1}{s}$ ways to order the remaining blocks. If it is 011, there are $\binom{k-1}{s-1}$ orders. Thus (1.6) follows. An equivalent form is

$$a_n = 2n \sum_{\substack{r, s \geq 0 \\ 2r + 3s = n}} \frac{2r + s}{r + s} \binom{r + s}{s}. \quad (5.2)$$

The equality of the summands follows from

$$2 \binom{k-1}{s} + \binom{k-1}{s-1} = \frac{2(k-s) + s}{k} \binom{k}{s}.$$

Although (5.2) contains a quotient, each summand inside the parentheses in (1.6) is visibly an integer. In particular, $a_n/(2n)$ is an integer, for a direct combinatorial reason.

One can also count the unmarked labeled cyclic words first. If there are k zeros, count pairs consisting of a word and one of its distinguished zeros. There are $n \binom{k}{s}$ such pairs, so the number of words is $\frac{n}{k} \binom{k}{s}$. Multiplication by $2r + s$ legal marks and by two pole choices recovers (5.2). This double count, unlike a naive orbit count, remains valid for periodic words.

5.2 Two useful recurrences

Corollary 5.1. *With $a_3 = 6$, $a_4 = 16$, $a_5 = 30$, one has*

$$(n-2)(n-3)a_n = n(n-3)a_{n-2} + n(n-2)a_{n-3}, \quad n \geq 6. \quad (5.3)$$

There is also an eventual constant-coefficient recurrence

$$a_n = 2a_{n-2} + 2a_{n-3} - a_{n-4} - 2a_{n-5} - a_{n-6}, \quad n \geq 9. \quad (5.4)$$

The order-six recurrence is of least possible order among eventual constant-coefficient linear recurrences for (a_n) .

Proof. Let $w_n = [x^n]W(x)$. From $QW = 2x^2 + x^3$ we obtain $w_n = w_{n-2} + w_{n-3}$ for $n \geq 4$. For $n \geq 6$ all three relevant sequence indices are at least three, and $a_m = 2mw_m$ gives (5.3) after clearing denominators.

Equation (5.4) follows by comparing coefficients in (4.3). Its numerator has degree eight, which is why the recurrence is asserted starting at $n = 9$, not earlier.

For minimality of the order, Q has only simple roots. Indeed, $Q' = -x(2 + 3x)$, while $Q(0) = 1$ and $Q(-2/3) = 23/27$ are nonzero. The numerator $N(x) = 2x^2 + x^3 = x^2(2 + x)$ of W has no common root with Q , because $Q(0) = 1$ and $Q(-2) = 5$. At a root α of Q , the numerator of $2xW'$ takes the nonzero value $-2\alpha N(\alpha)Q'(\alpha)$. Subtraction of the polynomial $8x^2$ cannot cancel the resulting double pole. Thus the reduced denominator of A is Q^2 , of degree six.

Any eventual constant-coefficient recurrence with polynomial $R(x)$ makes $R(x)A(x)$ a polynomial. The reduced denominator Q^2 must therefore divide R , forcing $\deg R \geq 6$. $\square$

The distinction is useful computationally: the three-step recurrence with polynomial coefficients does not contradict the minimal order-six statement, which is only about *constant* coefficients. The short recurrence for p_m gives a particularly simple exact algorithm and uses only integer additions before the final scaling.

6 Refinement by cardinality

Let $a_{n,d}$ count the sets of size d , and define

$$\mathcal{A}(x, y) = \sum_{n \geq 3} \sum_d a_{n,d} x^n y^d.$$

The encoding gives more than the total count without requiring a new graph-theoretic argument.

Theorem 6.1. *For $n \geq 3$, define*

$$r = 2n - 3d + 6, \quad s = 2d - n - 4, \quad k = n - d + 2. \quad (6.1)$$

If $r, s \geq 0$, then

$$a_{n,d} = \frac{2n(3n - 4d + 8)}{n - d + 2} \binom{n - d + 2}{2d - n - 4}. \quad (6.2)$$

Otherwise $a_{n,d} = 0$. Exactly the following cardinalities occur:

$$2 + \lceil n/2 \rceil \leq d \leq 2 + \lfloor 2n/3 \rfloor. \quad (6.3)$$

Moreover,

$$\mathcal{A}(x, y) = 2y^2 x \frac{\partial}{\partial x} \left(\frac{2x^2 y + x^3 y^2}{1 - x^2 y - x^3 y^2} \right) - 8x^2 y^3. \quad (6.4)$$

Proof. A block 01 contributes length two and one selected a -vertex; a block 011 contributes length three and two selected a -vertices. The two selected vertices outside these counts are the pole and the marked b -vertex. Solving

$$2r + 3s = n, \quad r + 2s = d - 2$$

gives (6.1). The fixed- (r, s) summand from (5.2) now gives (6.2), since $2r + s = 3n - 4d + 8$ and $r + s = k$. When $r, s \geq 0$, k is positive and the count is positive. Their nonnegativity is exactly the interval (6.3), including every integer in that interval.

For the generating function, unmarked blocks have weights x^2y and x^3y^2 , while the marked first block has weight $2x^2y + x^3y^2$. The factor y^2 records the selected pole and selected b -vertex. The derivative inserts the number of labeled positions, and the factor two allows both poles. The sole length-two term is $8x^2y^3$, so it is subtracted as before. $\square$

As a check, substituting $y = 1$ in (6.4) returns (4.2). For $n = 12$ the possible sizes are eight, nine, and ten, with

$$a_{12,8} = 48, \quad a_{12,9} = 384, \quad a_{12,10} = 24.$$

Their sum is $456 = a_{12}$. Thus inclusion-minimal dominating sets can have different cardinalities even after connectedness is imposed.

For direct symbolic use, expansion of (6.4) also gives

$$\mathcal{A}(x, y) = \frac{-2x^3y^4(4x^5y^3 + 8x^4y^2 + 4x^3y - 9x^2y - 8x - 3)}{(x^3y^2 + x^2y - 1)^2}. \quad (6.5)$$

The derivative form is preferable for explaining the combinatorics; the rational form is useful for automatic coefficient extraction.

7 A three-root formula and asymptotic behavior

7.1 Exact spectral form

Let $\lambda_1, \lambda_2, \lambda_3$ be the roots of

$$t^3 - t - 1 = 0. \quad (7.1)$$

They are distinct: a repeated root would also satisfy $3t^2 - 1 = 0$, and neither $t = 1/\sqrt{3}$ nor $t = -1/\sqrt{3}$ solves the cubic. Set

$$c_j = \frac{2\lambda_j + 1}{2\lambda_j + 3}. \quad (7.2)$$

Theorem 7.1. *For every $n \geq 3$,*

$$a_n = 2n \sum_{j=1}^3 c_j \lambda_j^n. \quad (7.3)$$

Proof. The roots $\alpha_j = 1/\lambda_j$ are the simple roots of $Q(x)$, and

$$Q(x) = \prod_{j=1}^3 (1 - \lambda_j x).$$

The rational function W has numerator and denominator of the same degree and polynomial part -1 . Its partial-fraction coefficient at $1 - \lambda_j x$ is

$$\begin{aligned} \lim_{x \rightarrow \alpha_j} (1 - \lambda_j x) W(x) &= -\frac{\lambda_j(2\alpha_j^2 + \alpha_j^3)}{Q'(\alpha_j)} \\ &= \frac{2 + \alpha_j}{2 + 3\alpha_j} = \frac{2\lambda_j + 1}{2\lambda_j + 3} = c_j. \end{aligned}$$

Therefore

$$W(x) = -1 + \sum_{j=1}^3 \frac{c_j}{1 - \lambda_j x}. \quad (7.4)$$

The polynomial part affects only the coefficient of x^0 . Extracting a positive-degree coefficient and using (1.3) proves the result. $\square$

7.2 The dominant root and an exponentially small absolute error

Let λ denote the positive real root of (7.1). This algebraic number is usually called the plastic constant; the name is not needed for the analysis. Its value and the leading constant below are

$$\lambda = 1.324717957244746025960908854478\dots, \quad (7.5)$$

$$C := \frac{2(2\lambda + 1)}{2\lambda + 3} = 1.291964709301166107055203050684\dots \quad (7.6)$$

These decimal values are computed in the supplied verifier; the exact definitions, rather than the decimals, are used in the following results.

Corollary 7.2. *As $n \rightarrow \infty$,*

$$\boxed{a_n = Cn\lambda^n + O(n\lambda^{-n/2})}. \quad (7.7)$$

In particular, $a_n \sim Cn\lambda^n$ and $a_{n+1}/a_n \rightarrow \lambda$.

Proof. The cubic has only one real root. Its local maximum, at $-1/\sqrt{3}$, is $2/(3\sqrt{3}) - 1 < 0$, and its local minimum is also negative. Since the value at one is negative and the value at two is positive, the real root satisfies $1 < \lambda < 2$.

Let $\mu, \bar{\mu}$ be the other two roots. Vieta's relations give

$$\mu + \bar{\mu} = -\lambda, \quad \mu\bar{\mu} = 1/\lambda,$$

so $|\mu| = \lambda^{-1/2} < 1$. Equation (7.3) becomes

$$a_n = Cn\lambda^n + 4n \operatorname{Re} \left(\frac{2\mu + 1}{2\mu + 3} \mu^n \right). \quad (7.8)$$

The second term is bounded in absolute value by a fixed constant times $n\lambda^{-n/2}$. This proves the asymptotic and its stated consequences. $\square$

The error in (7.7) is an *absolute* error tending to zero, not merely a small relative error. The two complex roots explain the small oscillation around the main term. In particular, there is no additional nonzero term of the form $C_0\lambda^n$: the dominant contribution in the exact formula is precisely a constant times $n\lambda^n$.

7.3 A uniform nearest-integer formula

Corollary 7.3. *For every integer $n \geq 45$,*

$$\boxed{a_n = \left\lfloor Cn\lambda^n + \frac{1}{2} \right\rfloor}. \quad (7.9)$$

The threshold 45 is sufficient; no minimality claim is made for it.

Proof. Let $c = (2\mu + 1)/(2\mu + 3)$. The Vieta relations used above imply

$$|c|^2 = \frac{4/\lambda - 2\lambda + 1}{4/\lambda - 6\lambda + 9} < 1.$$

The denominator is $|2\mu + 3|^2 > 0$, and it exceeds the numerator by $8 - 4\lambda > 0$. Thus (7.8) yields

$$|a_n - Cn\lambda^n| < 4n\lambda^{-n/2}.$$

The function $t^3 - t - 1$ is increasing for $t > 1$, and

$$(64/49)^3 - 64/49 - 1 = -\frac{9169}{117649} < 0.$$

Consequently $\lambda > 64/49$, so $\lambda^{-1/2} < 7/8$ and

$$|a_n - Cn\lambda^n| < 4n(7/8)^n.$$

The right side is decreasing for integer $n \geq 8$, and at $n = 45$ it is less than $1/2$. The latter is the exact integer inequality

$$360 \cdot 7^{45} < 8^{45}.$$

Hence the error is strictly less than $1/2$ for every $n \geq 45$. Since a_n is an integer, the rounding formula follows. $\square$

n	a_n	$a_n - Cn\lambda^n$ (approximately)
20	7160	2.701743849
45	18197730	-0.100362733
50	82488200	-0.083242473
100	210665294616400	0.000181710

These numerical values illustrate the exact theorem. A numerical rounding experiment alone would not justify a uniform statement for all subsequent indices.

8 Independent verification and reproducibility

The attached program `code/verify.py` requires only the Python standard library. Its principal exhaustive check deliberately does not assume the classification or use the private-neighbor lemma as a shortcut. It constructs the graph from (2.1), visits all vertex subsets, and applies the literal tests in the definition: coverage of all vertices, failure of every one-vertex deletion to dominate, and induced connectedness.

For the coverage test, vertices are represented by bits and closed neighborhoods by bit masks. If d is one set bit of a mask D , then

$$\text{coverage}(D) = \text{coverage}(D \setminus \{d\}) \text{ OR } N[d].$$

Processing masks in increasing order makes every smaller coverage mask available. Connectivity is tested by a separate graph search restricted to the selected vertices. A second routine generates sets from the cyclic binary word and mark description. The program compares the resulting *sets of subsets*, not just their sizes.

n	Vertices	Subsets examined	Count a_n
3	8	256	6
4	10	1024	16
5	12	4096	30
6	14	16384	36
7	16	65536	70
8	18	262144	96
9	20	1048576	144
10	22	4194304	220

For every row, exact subset families and their full cardinality distributions agree. The archive also contains private-neighbor certificates for all 30 sets when $n = 5$.

Additional tests use different algebraic representations. The program expands the conjectured rational function by exact formal division; computes the Padovan-type recurrence and binomial sum; checks both recurrences with their correct starting indices; and checks that summing (6.2) gives the total. These agree for every $3 \leq n \leq 1000$. All 40 published OEIS values, covering $3 \leq n \leq 42$, agree exactly with the result. The source values are retained with attribution in `data/oeis_a381190_3_42.txt`.

The exact rational inequalities in the nearest-integer proof are checked with integer arithmetic. Separately, a high-precision `Decimal` calculation checks the numerical rounding formula for $45 \leq n \leq 1000$. These decimal checks are diagnostics, not interval-arithmetic certificates; the uniform analytic proof is Corollary 7.3.

The default reproduction command, run from the archive root, is

```
python3 code/verify.py --brute-max 10 --max-n 1000
```

Use ordinary Python execution, not `python -O`, because the verifier uses assertions. The exhaustive part is exponential: T_n has $2n + 2$ vertices and 2^{2n+2} subsets. By contrast, the recurrence computes the first N sequence values with $O(N)$ integer arithmetic operations. These are arithmetic-operation counts, not unit-cost bit-complexity claims for unbounded integers.

A single term can also be obtained from a third-order recurrence by binary powering of its companion matrix in $O(\log n)$ integer matrix multiplications. This observation is an algorithmic consequence of (5.1); it is not needed by the verifier.

8.1 What the computation does and does not establish

The exhaustive checks are strong guards against a mistaken graph model, a reversed minimality convention, a missing small case, an incorrect symmetry factor, or an erroneous size refinement. They do not prove a statement about infinitely many graphs. That role belongs to [theorem 3.5](#) and the marked-block bijection. Likewise, agreement with all current OEIS terms is a useful external cross-check, not the logical basis of the generating function.

The delivered proof is a mathematical argument in ordinary language, accompanied by executable tests. It is not represented as machine-checked in Lean or another proof assistant. The source status and proposed attribution are documented; neither an OEIS edit nor a literature-priority claim is implied.

9 Conclusion

The conjecture is true. Connected minimal dominating sets of trapezohedral graphs admit a rigid global structure: exactly one pole is selected, and minimality forces a unique selected vertex of the opposite rim type. The rest of the problem becomes a cyclic word with runs of one or two ones, together with a legal marked edge. Distinguishing the marked block turns the cyclic enumeration into a linear block sequence without symmetry ambiguities.

This structure proves Arndt’s generating function and explains its squared cubic denominator. It also gives explicit general terms, all cardinality refinements, the exact range of occurring sizes, and a spectral formula whose subdominant terms decay in absolute value. The resulting nearest-integer expression recovers every term from $n = 45$ onward using a single real algebraic number.

A The candidate list and random selection

The four candidates below were recorded in the stated order before the draw. Each entry explicitly labeled the relevant formula as conjectural when inspected. The list deliberately avoids prime number problems and congruence-based sequences. Cyclic indices in the selected graph model are ordinary graph notation, not a modular-arithmetic subject.

No.	OEIS	Candidate and rationale
1	A225114	Skew partitions with no empty rows or columns. Kurkov’s continued-fraction generating function, proposed September 3, 2024. This is a natural area-enumeration problem for skew Young diagrams, rather than an artificially defined sequence [4].
2	A244475	Fifth-largest distinct term in rows of Stern’s diatomic triangle. Heinz’s rational generating function, proposed June 20, 2022. The target concerns extremal structure in a classical recursively defined array [5].
3	A289587	Permutations avoiding 321 and a specified length-two mesh pattern. Scheuerle’s algebraic generating function, proposed December 23, 2025. The problem refines a classical permutation avoidance class by a nonlocal pattern restriction [6].
4	A381190	Connected minimal dominating sets in trapezohedral graphs. Arndt’s rational generating function, proposed January 7, 2026. This combines three standard graph-theoretic properties in a polyhedral family beginning with the cube [1, 2].

The exact target expressions for the unselected entries can be read at the cited OEIS pages; no result about them is used in the proof. Previously addressed entries A199085, A260306, A158415, and A277364 were excluded from this selection.

The draw used the Python standard-library call

```
secrets.randbelow(4)
```

exactly once. Its zero-based result was 3, selecting one-based candidate 4, A381190. The recorded reroll count is zero. The candidate order and result are retained in `data/candidates.json` and

`data/selection.json`. The selection script refuses to overwrite an existing recorded draw. The verification script checks the existing record and does not request new randomness.

The randomness came from the operating system and was not generated from a published deterministic seed. The record therefore documents the actual selection, not a seed-replayable experiment. Re-running the mathematical verifier reproduces all proof checks without repeating or altering that choice.

B Archive guide

Artifact	Contents
<code>article.tex</code> , <code>article.pdf</code>	Source and compiled article.
<code>README.md</code> , <code>Makefile</code>	Build and verification commands, requirements, and scope notes.
<code>code/verify.py</code>	Exact graph enumeration, structural generator, formula checks, and numerical diagnostics.
<code>code/select_candidate.py</code>	Single-draw selection utility with an overwrite guard.
<code>data/candidates.json</code>	Ordered pre-draw candidate list.
<code>data/selection.json</code>	Recorded selection and no-reroll audit.
<code>data/verification.txt</code>	Captured successful verification run.
<code>data/exhaustive_checks.json</code>	Counts and size distributions for the exhaustive cases.
<code>data/terms.csv</code>	Terms for $3 \leq n \leq 1000$.
<code>data/size_distribution.csv</code>	Counts by cardinality for $3 \leq n \leq 100$.
<code>data/certificates_n5.json</code>	All 30 sets for T_5 with private-neighbor witnesses.
<code>data/asymptotics.json</code>	High-precision constants and sample absolute errors.
<code>notes/sources.md</code>	Source URLs, retrieval date, and attribution notes.
<code>notes/oeis_submission.txt</code>	An unsubmitted draft of a concise proof explanation and additional formulas.

References

- [1] The OEIS Foundation Inc., *A381190: Number of connected minimal dominating sets in the n -trapezohedral graph*. Sequence by Eric W. Weisstein, February 16, 2025; extension by Christian Sievers, January 2, 2026; generating-function conjecture by Joerg Arndt, January 7, 2026. Internal revision 16, January 7, 2026. <https://oeis.org/A381190>; internal record: <https://oeis.org/A381190/internal>. Accessed September 20, 2026.
- [2] Eric W. Weisstein, *Trapezohedral Graph*, MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/TrapezohedralGraph.html>. Accessed September 20, 2026.
- [3] Eric W. Weisstein, *Minimal Dominating Set*, MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/MinimalDominatingSet.html>. Accessed September 20, 2026.
- [4] The OEIS Foundation Inc., *A225114: Number of skew partitions of n with no empty rows or columns*. Continued-fraction conjecture by Mikhail Kurkov, September 3, 2024. <https://oeis.org/A225114>. Accessed September 20, 2026.
- [5] The OEIS Foundation Inc., *A244475: Fifth largest term in the n -th row of Stern's diatomic triangle*. Rational generating-function conjecture by Alois P. Heinz, June 20, 2022. The entry's programs use distinct values for the rank. <https://oeis.org/A244475>. Accessed September 20, 2026.
- [6] The OEIS Foundation Inc., *A289587: Number of 321-avoiding permutations avoiding the mesh pattern $(12, 174)$* . Algebraic generating-function conjecture by Thomas Scheuerle, December 23, 2025. <https://oeis.org/A289587>. Accessed September 20, 2026.